

Exercise 01: Display The output (Html with inline CSS)

Column Span

Courses Table		
ID	code	Name
10	CSCI250	Introduction to Programming
20	CSCI300	Intermediate Programming with Objects
30	CSCI390	Web Development

Row Span

Courses Table	ID	code	Name
	10	CSCI250	Introduction to Programming
	20	CSCI300	Intermediate Programming with Objects
	30	CSCI390	Web Development

Cell Spacing - Cell Padding

ID	code	Name
10	CSCI250	Introduction to Programming
20	CSCI300	Intermediate Programming with Objects
30	CSCI390	Web Development

Non-Structural Group

ID	Name
10	Web Development

Structural Column Group

ID	Code	Name
10	CS101	Web Development

```
<!DOCTYPE html>
<html>
<head>
  <title>Tables Example</title>
</head>
<body style="font-family:Arial,sans-serif;">
<h1>Column Span</h1>
<table border="1" style="border-collapse:collapse;border:2px solid green;width:800px;">
  <tr>
    <th colspan="3" style="border:2px solid green;padding:8px;">Courses Table</th>
  </tr>
  <tr>
    <th style="border:2px solid green;padding:8px;">ID</th>
    <th style="border:2px solid green;padding:8px;">code</th>
    <th style="border:2px solid green;padding:8px;">Name</th>
  </tr>
  <tr>
    <td style="border:2px solid green;padding:8px;">10</td>
    <td style="border:2px solid green;padding:8px;">CSCI250</td>
    <td style="border:2px solid green;padding:8px;">Introduction to Programming</td>
  </tr>
  <tr>
    <td style="border:2px solid green;padding:8px;">20</td>
    <td style="border:2px solid green;padding:8px;">CSCI300</td>
    <td style="border:2px solid green;padding:8px;">Intermediate Programming with Objects</td>
  </tr>
</table>
```

```

<tr>
  <td style="border:2px solid green;padding:8px;">30</td>
  <td style="border:2px solid green;padding:8px;">CSCI390</td>
  <td style="border:2px solid green;padding:8px;">Web Development</td>
</tr>
</table>
<br><br>
<h1>Row Span</h1>
<table border="1" style="border-collapse:collapse;border:2px solid green;width:800px;">
  <tr>
    <td rowspan="4" style="border:2px solid green;padding:8px;width:150px;vertical-align:top;">
      Courses<br>Table
    </td>
    <th style="border:2px solid green;padding:8px;">ID</th>
    <th style="border:2px solid green;padding:8px;">code</th>
    <th style="border:2px solid green;padding:8px;">Name</th>
  </tr>
  <tr>
    <td style="border:2px solid green;padding:8px;">10</td>
    <td style="border:2px solid green;padding:8px;">CSCI250</td>
    <td style="border:2px solid green;padding:8px;">Introduction to Programming</td>
  </tr>
  <tr>
    <td style="border:2px solid green;padding:8px;">20</td>
    <td style="border:2px solid green;padding:8px;">CSCI300</td>
    <td style="border:2px solid green;padding:8px;">Intermediate Programming with Objects</td>
  </tr>
  <tr>
    <td style="border:2px solid green;padding:8px;">30</td>
    <td style="border:2px solid green;padding:8px;">CSCI390</td>
    <td style="border:2px solid green;padding:8px;">Web Development</td>
  </tr>
</table>
<br><br>
<h1>Cell Spacing - Cell Padding</h1>
<table border="1" cellspacing="15" cellpadding="15" style="border:2px solid green;width:800px;">
  <tr style="background:gray;">
    <th>ID</th>
    <th>code</th>
    <th>Name</th>
  </tr>
  <tr>
    <td>10</td>
    <td>CSCI250</td>
    <td>Introduction to Programming</td>
  </tr>
  <tr>
    <td>20</td>
    <td>CSCI300</td>
    <td>Intermediate Programming with Objects</td>
  </tr>
  <tr>
    <td>30</td>
    <td>CSCI390</td>
    <td>Web Development</td>
  </tr>
</table>
<br><br>
<h1>Non-Structural Group</h1>
<table border="1" style="border-collapse:collapse;width:640px;">
  <colgroup>
    <col style="background:#dddddd;">
    <col style="background:#bfe8ff;">

```

```

</colgroup>
<tr>
  <th style="padding:8px;">ID</th>
  <th style="padding:8px;">Name</th>
</tr>
<tr>
  <td style="padding:8px;">10</td>
  <td style="padding:8px;">Web Development</td>
</tr>
</table>
<br><br>
<h1>Structural Column Group</h1>
<table border="1"
  style="border-collapse:collapse;width:640px;">
  <colgroup span="2"></colgroup>
  <colgroup></colgroup>
  <tr>
    <th style="padding:8px;">ID</th>
    <th style="padding:8px;">Code</th>
    <th style="padding:8px;">Name</th>
  </tr>
  <tr>
    <td style="padding:8px;">10</td>
    <td style="padding:8px;">CS101</td>
    <td style="padding:8px;">Web Development</td>
  </tr>
</table>
</body>
</html>

```

Exercise 02 : Display This Output (Html)

First Name

Last Name

Age

Date of Birth

Favorite Colors: ☐ Green ☐ Red ☐ Yellow ☐ Violet ☐ Pink

Favorite Fruite: ☐ Apple ☐ Bananna ☐ Watermelon ☐ Appricot ☐ Orange

City

Comments

```

<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
</head>
<body style="font-family:Arial; font-size:20px;">
<form>
<table>
<tr>
  <td style="padding:8px;">First Name</td>
  <td><input type="text" style="width:220px;"></td>
</tr>
<tr>
  <td style="padding:8px;">Last Name</td>
  <td><input type="text" style="width:380px;"></td>
</tr>
<tr>
  <td style="padding:8px;">Age</td>
  <td><input type="number" style="width:380px;"></td>
</tr>
<tr>
  <td style="padding:8px;">Date of Birth</td>
  <td>
    <select style="width:280px;">
      <option></option>
      <option>01/01/2000</option>
      <option>02/02/2001</option>
      <option>03/03/2002</option>
    </select>
  </td>
</tr>
<tr>
  <td style="padding:8px;">Favorite Colors:</td>
  <td>
    <input type="checkbox"> Green
    <input type="checkbox"> Red
    <input type="checkbox"> Yellow
    <input type="checkbox"> Violet
    <input type="checkbox"> Pink
  </td>
</tr>
<tr>
  <td style="padding:8px;">Favorite Fruite:</td>
  <td>
    <input type="radio" name="fruit"> Apple
    <input type="radio" name="fruit"> Bananna
    <input type="radio" name="fruit"> Watermelon
    <input type="radio" name="fruit"> Appricot
    <input type="radio" name="fruit"> Orange
  </td>
</tr>
<tr>
  <td style="padding:8px;">City</td>
  <td>
    <select size="4" style="width:120px;">
      <option>Beirut</option>
      <option>Nabatieh</option>
      <option>Tyre</option>
      <option>Saida</option>
    </select>
  </td>
</tr>
<tr>

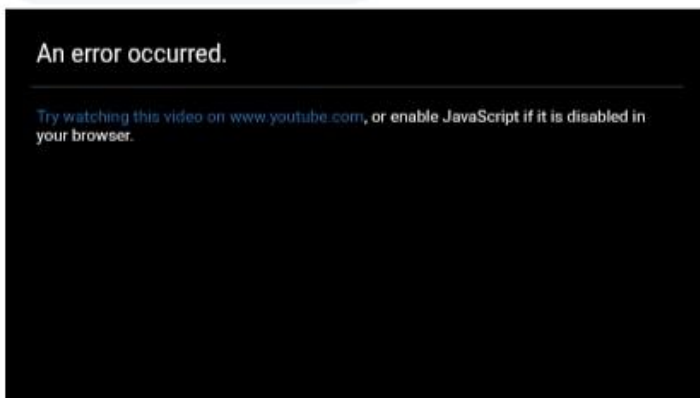
```

```

<td style="padding:8px; vertical-align:top;">Comments</td>
<td>
  <textarea rows="6" cols="65"></textarea>
</td>
</tr>
<tr>
<td></td>
<td>
  <input type="submit" value="Register">
  <input type="reset" value="Clear">
</td>
</tr>
</table>
</form>
</body>
</html>

```

Exercise 03 : Display The Output (Html)



```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Media Example</title>
<style>
  body{
    font-family: Arial, sans-serif;
    background:#fff;
    padding:10px;
  }
  video{
    width:450px;
    display:block;
    margin-bottom:15px;
  }
  audio{
    width:420px;
    display:block;
    margin-bottom:20px;
  }
  iframe{
    width:780px;
    height:440px;
    border:none;
  }
</style>
</head>
<body>
<!-- Video -->
<video controls poster="https://via.placeholder.com/450x330">
  <source src="video.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
<!-- Audio -->
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
<!-- YouTube Embed -->
<iframe src="https://www.youtube.com/embed/INVALID_VIDEO_ID" allowfullscreen>
</iframe>
</body>
</html>

```

Exercise 04 : Write the Output (Html and CSS)



testo4.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Layout Example</title>
  <link rel="stylesheet" href="testo4.css">
</head>
<body>
<div class="box">Div 1</div>
<div class="box">Div 2</div>
<p class="box">This is my paragraph</p>
<p class="box">This is another paragraph</p>
<section class="section-red">
  <h2>My Section</h2>
</section>
<section>
  <h2>Another Section</h2>
</section>
<article class="article-box">
  <h2>Article 1</h2>
</article>
<article>
  <h2>Article 2</h2>
</article>
```

```

<header class="article-box">
  <h2>My Header</h2>
</header>
<header>
  <h2>Another Header</h2>
</header>
<button class="green-btn">Save</button>
<button class="green-btn">cancel</button>
<button class="green-btn red-text">cancel</button>
<div class="button-container">
  <button class="blue-btn">Button 1</button>
</div>
</body>
</html>

```

testo4.css:

```

body{
  font-family: Arial, sans-serif;
  margin:10px;
}
.box{
  background:#19dbe2;
  padding:18px;
  margin:12px 0;
  border-radius:10px;
  font-size:22px;
  font-weight:bold;
}
.section-red{
  border:4px solid red;
  padding:20px;
  margin:12px 0;
}
.article-box{
  border:4px solid green;
  border-radius:15px;
  padding:25px;
  margin:12px;
}
.green-btn{
  width:100%;
  background:green;
  color:white;
  border:none;
  padding:18px;
  margin:12px 0;
  border-radius:10px;
  font-size:22px;
  font-weight:bold;
  cursor:pointer;
}
.red-text{
  color:red;
}
.button-container{
  background:#19dbe2;
  border:4px solid green;
  border-radius:12px;
  padding:35px;
  margin-top:15px;
}
.blue-btn{
  width:100%;

```

```
background:blue;
color:white;
border:none;
border-radius:12px;
padding:18px;
font-size:22px;
font-weight:bold;
cursor:pointer;
}
```

Exercise 05: Display The Output (Html / CSS / JavaScript)

Question 1 [25 points]:

Write the following HTML code:

Product Cost Calculator

Item-1 Price	Item-2 Price	Item-3 Price
Calculate		
Total:		

Create a JS function "calculateTotal()":

1. Name the main variable using your first, middle, or last name (e.g., haniTotal).
2. The user fills textboxes: "Item-1", "Item-2" and "Item-3" and presses the button. If any textbox is empty, show an alert: "You must fill all price fields."
3. If all values are filled:
 - 3.a. Calculate the sum. If sum ≥ 200 , apply a 15% discount. Display the new cost in the result textbox. Show a red message below the button stating the discount applied and final total.
 - 3.b. Otherwise, display the exact sum in the textbox and show a message below stating no discount was offered.

Product Cost Calculator

Calculate

Total:

No discount offered. Total: \$60.00

Product Cost Calculator

Calculate

Total:

15% Discount Applied! New Total: \$280.50

Question 2 [25 points]:

Write the following HTML code:

Quick Order Tracker

Customer	Items	Total (\$25/each)
----------	-------	-------------------

Create a JS function "addOrder()":

1. Name the tracking variable using your first, middle, or last name, ex: alexOrders.
2. User enters "Customer Name" and "Number of Items", then clicks the button (Assume a fixed price of 25\$ per item).
3. When mouse moves over the table, border changes to dashed orange and width to 600px.
4. When mouse moves out, border resets to solid gray and width changes to 450px.
5. On button click, validate inputs, if valid, calculate the line total (items * 25\$), add a new row to the table showing Name, items and Total.
6. Update a paragraph below the button showing total orders placed and total revenue.

Quick Order Tracker

Total Orders: 2 | Total Revenue: \$175.00

Customer	Items	Total (\$25/each)
Mazen	3	\$75.00
Fadi	4	\$100.00

question 1:

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Product Cost Calculator</title>
<style>
body{
  font-family: Arial;
  margin:40px;
}
input{
  width:180px;
  padding:10px;
  margin:10px;
}
button{
  padding:10px 20px;
  margin:10px;
}
#msg{
  font-size:20px;
  font-weight:bold;
}
</style>
<script>
function calculateTotal(){
  let jawadTotal = 0;
  let item1 = document.getElementById("item1").value;
  let item2 = document.getElementById("item2").value;
  let item3 = document.getElementById("item3").value;
  if(item1=="" || item2=="" || item3==""){
    alert("You must fill all price fields.");
    return;
  }
  jawadTotal = Number(item1)+Number(item2)+Number(item3);
  if(jawadTotal>=200){
    jawadTotal = jawadTotal*0.85;
    document.getElementById("total").value=jawadTotal.toFixed(2);
    document.getElementById("msg").style.color="red";
    document.getElementById("msg").innerHTML=
    "15% Discount Applied! New Total: $" + jawadTotal.toFixed(2);
  }
  else{
    document.getElementById("total").value=jawadTotal.toFixed(2);
    document.getElementById("msg").style.color="green";
    document.getElementById("msg").innerHTML=
    "No discount offered. Total: $" + jawadTotal.toFixed(2);
  }
}
</script>
</head>
<body>
<h1>Product Cost Calculator</h1>
<input type="text" id="item1" placeholder="Item-1 Price">
<input type="text" id="item2" placeholder="Item-2 Price">
<input type="text" id="item3" placeholder="Item-3 Price">
<br><br>
<button onclick="calculateTotal()">Calculate</button>
<br><br>
<label>Total:</label>
<input type="text" id="total" readonly>
<p id="msg"></p>
```

```
</body>
</html>
```

question 2:

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Quick Order Tracker</title>
<style>
body{
  font-family:Arial;
  margin:40px;
}
input{
  padding:10px;
  width:200px;
  margin:10px;
}
button{
  padding:10px 20px;
}
table{
  width:450px;
  border-collapse:collapse;
  border:2px solid gray;
}
th,td{
  border:1px solid gray;
  padding:10px;
  text-align:center;
}
</style>
<script>
let jawadOrders = 0;
let totalRevenue = 0;
function addOrder(){
  let name=document.getElementById("name").value;
  let items=document.getElementById("items").value;
  if(name=="" || items==""){
    alert("Fill all fields");
    return;
  }
  let total=Number(items)*25;
  jawadOrders++;
  totalRevenue+=total;
  let table=document.getElementById("orders");
  let row=table.insertRow(-1);
  row.insertCell(0).innerHTML=name;
  row.insertCell(1).innerHTML=items;
  row.insertCell(2).innerHTML="$"+total.toFixed(2);
  document.getElementById("info").innerHTML=
  "Total Orders: "+jawadOrders+
  " | Total Revenue: $" +totalRevenue.toFixed(2);
}
function mouseIn(){
  let table=document.getElementById("orders");
  table.style.border="3px dashed orange";
  table.style.width="600px";
}
function mouseOut(){
  let table=document.getElementById("orders");
  table.style.border="2px solid gray";
}
```

```

    table.style.width="450px";
  }
</script>
</head>
<body>
<h1>Quick Order Tracker</h1>
<input type="text" id="name" placeholder="Customer Name">
<input type="number" id="items" placeholder="Number of Items">
<button onclick="addOrder()">Add Order</button>
<p id="info">
Total Orders: 0 | Total Revenue: $0.00
</p>
<table id="orders" onmouseover="mouseIn()" onmouseout="mouseOut()">
<tr>
<th>Customer</th>
<th>Items</th>
<th>Total ($25/each)</th>
</tr>
</table>
</body>
</html>

```

Exercise 06 : React

Instructions

1. Use the created folder of the exam
2. No new dependencies need to be installed.

Output:



Task 1 – Project Setup (10 pts)

- Install React successfully.
- Show a simple home page with the title:
Welcome to Book Explorer

Task 2 – Navbar Component (15 pts)

- Create **Navbar.js** inside **src/components**.
- The Navbar should have links: **Home, Books, Contact**.
- Use **React Router** for navigation.
- The active link should visually stand out (e.g. bold or colored).

Task 3 – Components & Props (20 pts)

- In **src/pages/Books.js**, create a functional component **BookCard** that receives props:
 - o title
 - o author
- Create a hardcoded array of books and render them using **map()** with these data:
 - o title: "The React Journey", author: "Chris Minnick"
 - o title: "Learning Node", author: "Shelley Powers"
 - o title: "JavaScript Magic", author: "Kyle Simpson"

- Display each book card nicely with title and author.



Task 4 – useState & Event Handling (20 pts)

- In the Navbar, add a button labeled **Toggle Welcome**.
- When clicked, it should show/hide the "Welcome to Book Explorer" title on the Home page using **useState**.
- **Bonus (+5 pts):** Change the button text dynamically between *Show Welcome / Hide Welcome*.

Task 5 – Form Handling (25 pts)

- In **src/pages/Contact.js**, create a simple contact form with:
 - o Full Name
 - o Email
 - o Message
- Manage form values with **useState**.
- On submit:
 - o If all fields are filled → show success message:
"Thank you {Name}, your message has been received."
 - o If any field is empty → show error message:
"Please fill all fields before submitting."



Task 6 – Styling and Structure (10 pts)

- Use CSS files or inline styles for a neat, consistent layout.
- Maintain a consistent color theme across all pages.

File Structure:

```
book-explorer/
├── src/
│   ├── components/
│   │   ├── Navbar.js
│   │   └── Navbar.css
│   ├── pages/
│   │   ├── Home.jsx
│   │   ├── Books.jsx
│   │   ├── Contact.jsx
│   │   ├── Home.css
│   │   ├── Books.css
│   │   └── Contact.css
│   └── App.js
```

```
| |— App.css
| |— index.js
| |— index.css
|— package.json
```

package.json:

```
{
  "name": "book-explorer",
  "version": "1.0.0",
  "private": true,
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "react-router-dom": "^6.22.3",
    "react-scripts": "5.0.1"
  },
  "scripts": {
    "start": "react-scripts start",
    "build": "react-scripts build"
  }
}
```

src/index.js:

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App";
import "./index.css";
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);
```

App.jsx:

```
import { Routes, Route } from "react-router-dom";
import { useState } from "react";
import Navbar from "./components/Navbar";
import Home from "./pages/Home";
import Books from "./pages/Books";
import Contact from "./pages/Contact";
import "./App.css";
function App() {
  const [showWelcome, setShowWelcome] = useState(true);
  return (
    <>
      <Navbar
        showWelcome={showWelcome}
        setShowWelcome={setShowWelcome}
      />
      <Routes>
        <Route
          path="/"
          element={<Home showWelcome={showWelcome} />}
        />
        <Route path="/books" element={<Books />} />
        <Route path="/contact" element={<Contact />} />
      </Routes>
    </>
  );
}
```

```
}  
export default App;
```

Navbar.jsx:

```
import { NavLink } from "react-router-dom";  
import "./Navbar.css";  
function Navbar({ showWelcome, setShowWelcome }) {  
  return (  
    <nav className="navbar">  
      <NavLink  
        to="/"  
        className={({ isActive }) => (isActive ? "active" : "")}>  
        >  
        Home  
      </NavLink>  
      <NavLink  
        to="/books"  
        className={({ isActive }) => (isActive ? "active" : "")}>  
        >  
        Books  
      </NavLink>  
      <NavLink  
        to="/contact"  
        className={({ isActive }) => (isActive ? "active" : "")}>  
        >  
        Contact  
      </NavLink>  
      <button onClick={() => setShowWelcome(!showWelcome)}>  
        {showWelcome ? "Hide Welcome" : "Show Welcome"}  
      </button>  
    </nav>  
  );  
}
```

```
export default Navbar;
```

Home.jsx:

```
import "./Home.css";  
function Home({ showWelcome }) {  
  return (  
    <div className="container">  
      {showWelcome && <h1>Welcome to Book Explorer</h1>}  
    </div>  
  );  
}
```

```
export default Home;
```

Books.jsx:

```
import "./Books.css";  
function BookCard({ title, author }) {  
  return (  
    <div className="card">  
      <h2>{title}</h2>  
      <p>{author}</p>  
    </div>  
  );  
}  
function Books() {  
  const books = [  
    {  
      title: "The React Journey",  
      author: "Chris Minnick"  
    },  
  ],
```

```

    {
      title: "Learning Node",
      author: "Shelley Powers"
    },
    {
      title: "JavaScript Magic",
      author: "Kyle Simpson"
    }
  ];
  return (
    <div className="container">
      <h1>Books</h1>
      {books.map((book, index) => (
        <BookCard
          key={index}
          title={book.title}
          author={book.author}
        />
      ))}
    </div>
  );
}
export default Books;

```

Contact.jsx:

```

import { useState } from "react";
import "../Contact.css";
function Contact() {
  const [fullName, setFullName] = useState("");
  const [email, setEmail] = useState("");
  const [message, setMessage] = useState("");
  const [result, setResult] = useState("");
  const handleSubmit = (e) => {
    e.preventDefault();
    if (fullName && email && message) {
      setResult(
        `Thank you ${fullName}, your message has been received.`
      );
    } else {
      setResult(
        "Please fill all fields before submitting."
      );
    }
  };
  return (
    <div className="container">
      <h1>Contact</h1>
      <form onSubmit={handleSubmit}>
        <input
          type="text"
          placeholder="Full Name"
          value={fullName}
          onChange={(e) => setFullName(e.target.value)}
        />
        <input
          type="email"
          placeholder="Email"
          value={email}
          onChange={(e) => setEmail(e.target.value)}
        />
        <textarea

```

```

        placeholder="Message"
        value={message}
        onChange={(e) => setMessage(e.target.value)}
      />
      <button type="submit">
        Submit
      </button>
    </form>
    <p>{result}</p>
  </div>
);
}
export default Contact;

```

Navbar.css:

```

.navbar{
  background:#1e3a8a;
  padding:15px;
  display:flex;
  gap:20px;
  align-items:center;
}
.navbar a{
  color:white;
  text-decoration:none;
}
.active{
  font-weight:bold;
  color:yellow;
}
button{
  margin-left:auto;
  padding:8px 15px;
}
src/pages/Home.css
.container{
  text-align:center;
  padding:30px;
}

```

Books.css:

```

.container{
  padding:30px;
}
.card{
  border:1px solid #ccc;
  padding:15px;
  margin:15px 0;
  border-radius:8px;
}

```

Contact.css:

```

.container{
  padding:30px;
}
form{
  display:flex;
  flex-direction:column;
  gap:15px;
  width:350px;
}
input,

```

```
textarea{
  padding:10px;
}
button{
  width:150px;
}
```

App.css:

```
body{
  margin:0;
  font-family:Arial, sans-serif;
  background:#f4f4f4;
}
```

index.css:

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
}
```

Exercise 07 : Build a Responsive University Website

Create a responsive website called University Portal using HTML, CSS, and Bootstrap 5.

Part 1 – Header (20 Marks):

Create a responsive Bootstrap Navbar.

Add the website logo "University Portal".

Add the following navigation links:

Home

Courses

Teachers

Contact

Include a Bootstrap Search Form on the right.

The Navbar should collapse into a hamburger menu on small screens.

Part 2 – Hero Section (20 Marks):

Create a Hero section using Bootstrap.

Requirements:

Large heading: Welcome to University Portal

Short paragraph describing the website.

Two Bootstrap buttons:

Explore Courses

Contact Us

Add an image beside the text using the Bootstrap Grid System.

Part 3 – Courses Section (20 Marks):

Create a section titled Our Courses.

Display 6 Bootstrap Cards.

Each card should contain:

Course Image

Course Name

Short Description

Bootstrap "Read More" Button

Arrange the cards using the Bootstrap Grid System so they are responsive.

Part 4 – Statistics Section (10 Marks):

Create a Bootstrap section displaying:

2500+ Students

120 Teachers

75 Courses

15 Years Experience

Use Bootstrap Columns.

Part 5 – Contact Form (20 Marks):

Create a Bootstrap Form that contains:

Full Name

Email

Phone Number

Subject

Message

Submit Button

Use Bootstrap Form Controls and proper spacing.

Part 6 – Footer (10 Marks):

Create a Bootstrap Footer containing:

Copyright © 2026 University Portal

Facebook icon

Instagram icon

Twitter icon

LinkedIn icon

Use Bootstrap Icons and align everything in the center.

index.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>University Portal</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-
icons.min.css">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">
<a class="navbar-brand" href="#"> University Portal </a>
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#menu">
<span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="menu">
<ul class="navbar-nav ms-auto">
<li class="nav-item">
<a class="nav-link" href="#">Home</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Courses</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Teachers</a>
</li>
<li class="nav-item">
```

```

<a class="nav-link" href="#">Contact</a>
</li>
</ul>
<form class="d-flex ms-3">
<input class="form-control me-2" type="search" placeholder="Search">
<button class="btn btn-primary"> Search </button>
</form>
</div>
</div>
</nav>
<section class="container mt-5">
<div class="row align-items-center">
<div class="col-md-6">
<h1>Welcome to University Portal</h1>
<p>Learn programming, web development, AI, networking and many other courses. </p>
<button class="btn btn-primary"> Explore Courses </button>
<button class="btn btn-outline-dark"> Contact Us </button>
</div>
<div class="col-md-6"> </div>
</div>
</section>
<section class="container mt-5">
<h2 class="text-center mb-4"> Our Courses </h2>
<div class="row">
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Web Development </h5>
<p class="card-text"> Learn HTML, CSS, JavaScript and Bootstrap. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Python </h5>
<p class="card-text"> Programming with Python from beginner to advanced. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Artificial Intelligence </h5>
<p class="card-text"> Introduction to AI and Machine Learning. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Networking </h5>
<p class="card-text"> Learn computer networking fundamentals. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>

```

```

</div>
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Database </h5>
<p class="card-text"> Learn SQL and database management. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>
<div class="col-md-4 mb-4">
<div class="card h-100">

<div class="card-body">
<h5 class="card-title"> Cyber Security </h5>
<p class="card-text"> Protect systems and networks from attacks. </p>
<a href="#" class="btn btn-primary"> Read More </a>
</div>
</div>
</div>
</div>
</div>
</section>
<section class="container text-center mt-5">
<div class="row">
<div class="col-md-3">
<h2>2500+</h2>
<p>Students</p>
</div>
<div class="col-md-3">
<h2>120</h2>
<p>Teachers</p>
</div>
<div class="col-md-3">
<h2>75</h2>
<p>Courses</p>
</div>
<div class="col-md-3">
<h2>15</h2>
<p>Years Experience</p>
</div>
</div>
</div>
</section>
<section class="container mt-5 mb-5">
<h2 class="text-center mb-4"> Contact Us </h2>
<form>
<div class="row">
<div class="col-md-6 mb-3">
<input type="text" class="form-control" placeholder="Full Name">
</div>
<div class="col-md-6 mb-3">
<input type="email" class="form-control" placeholder="Email">
</div>
</div>
<div class="row">
<div class="col-md-6 mb-3">
<input
type="text" class="form-control" placeholder="Phone Number">
</div>
<div class="col-md-6 mb-3">
<input type="text" class="form-control" placeholder="Subject">
</div>
</div>
</div>

```

```

<div class="mb-3">
  <textarea class="form-control" rows="5" placeholder="Message">
</textarea>
</div>
<button class="btn btn-success"> Submit </button>
</form>
</section>
<footer class="bg-dark text-white text-center p-4">
<p> Copyright © 2026 University Portal </p>
<i class="bi bi-facebook fs-3 m-2"></i>
<i class="bi bi-instagram fs-3 m-2"></i>
<i class="bi bi-twitter fs-3 m-2"></i>
<i class="bi bi-linkedin fs-3 m-2"></i>
</footer>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

style.css:

```

body{
  background:#f8f9fa;
}
h1,h2{
  font-weight:bold;
}
.navbar-brand{
  font-size:28px;
  font-weight:bold;
}
.card{
  transition:0.3s;
}
.card:hover{
  transform:scale(1.05);
  box-shadow:0 0 15px gray;
}
.card img{
  height:220px;
  object-fit:cover;
}
.btn{
  margin-top:10px;
}
footer i{
  cursor:pointer;
  transition:0.3s;
}
footer i:hover{
  color:gold;
}
section{
  padding-top:30px;
  padding-bottom:30px;
}
textarea{
  resize:none;
}

```

Exercise 08 :PhP

Create a simple Book Order website connected to a PHP backend.

Task 1 (10 Marks):

Create an HTML page named index.html that contains a form with the following fields:

Customer Name

Email

Book Title

Quantity

Submit Button

The form must send data to order.php using the POST method.

Task 2 (20 Marks):

Create order.php that:

Receives the submitted data.

Stores the values into PHP variables.

Displays all submitted information.

Task 3 (20 Marks):

Validate the form:

If any field is empty, display:

Please fill all fields.

Otherwise continue.

Task 4 (20 Marks):

Calculate the total price.

Book price:

£15

Formula:

Total = Quantity × 15

Display the total price.

Task 5 (15 Marks):

If Quantity is greater than or equal to 3, display:

You received a 10% discount.

Also display the final price after discount.

Task 6 (15 Marks):

Use CSS to make the form look clean and centred.

```
<!DOCTYPE html>
<html>
<head>
<!DOCTYPE html>
<html>
<head>
  <title>Book Order</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>Book Order Form</h1>
<form action="order.php" method="POST">
<input type="text" name="name" placeholder="Customer Name">
<input type="email" name="email" placeholder="Email">
<input type="text" name="book" placeholder="Book Title">
<input type="number" name="quantity" placeholder="Quantity">
```

```
<button type="submit"> Order </button>
</form>
</div>
</body>
</html>
```

order.php:

```
<?php $name = $_POST["name"]; $email = $_POST["email"]; $book = $_POST["book"];
$quantity = $_POST["quantity"];
if( empty($name) || empty($email) || empty($book) || empty($quantity) ){
    echo "<h2>Please fill all fields.</h2>";
}
else{
    $price = 15;
    $total = $quantity * $price;
    echo "<h2>Order Details</h2>";
    echo "Customer Name: $name <br>";
    echo "Email: $email <br>";
    echo "Book Title: $book <br>";
    echo "Quantity: $quantity <br>";
    echo "Total Price: £$total <br>";
    if($quantity >= 3){
        $discount = $total * 0.10;
        $finalPrice = $total - $discount;
        echo "<br>";
        echo "You received a 10% discount.<br>";
        echo "Final Price: £$finalPrice";
    }
}
?>
```

style.css:

```
body{
    font-family: Arial;
    background:#f2f2f2;
}
.container{
    width:400px;
    margin:50px auto;
    background:white;
    padding:20px;
    border-radius:10px;
}
input{
    width:100%;
    padding:10px;
    margin:10px 0;
}
button{
    width:100%;
    padding:10px;
    background:green;
    color:white;
    border:none;
}
```

Exercise 09 : Node.js

Create a Student Management System using React.js, Node.js, Express.js, and MySQL.

Part 1 – Database (20 Marks):

Create a MySQL database called student_db.

Create a table called students with the following columns:

id (INT, AUTO_INCREMENT, PRIMARY KEY)

name (VARCHAR(100))

email (VARCHAR(100))

major (VARCHAR(100))

Insert at least 3 sample students into the table.

Part 2 – Backend (40 Marks):

Using Node.js and Express.js, connect your application to MySQL.

Create the following REST API endpoints:

GET /students → Return all students.

GET /students/:id → Return one student by ID.

POST /students → Add a new student.

PUT /students/:id → Update a student.

DELETE /students/:id → Delete a student.

All requests and responses should use JSON.

Part 3 – Frontend (40 Marks):

Using React.js, create a page that:

Displays all students in a table.

Includes a form to add a new student.

Allows updating an existing student.

Allows deleting a student.

Uses Axios to communicate with the backend.

Database (MySQL)

```
CREATE DATABASE student_db;
USE student_db;
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100),
  major VARCHAR(100)
);
INSERT INTO students(name,email,major)
VALUES
('John Smith','john@gmail.com','Computer Science'),
('Sara Ali','sara@gmail.com','Information Technology'),
('Mike Brown','mike@gmail.com','Software Engineering');
```

server.js:

```
const express = require("express");
const mysql = require("mysql2");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "",
  database: "student_db"
});
```

```

db.connect(err => {
  if(err) throw err;
  console.log("MySQL Connected");
});
app.get("/students", (req, res) => {
  db.query("SELECT * FROM students", (err, result) => {
    if(err) throw err;
    res.json(result);
  });
});
app.get("/students/:id", (req, res) => {
  db.query(
    "SELECT * FROM students WHERE id=?",
    [req.params.id],
    (err, result) => {
      if(err) throw err;
      res.json(result);
    }
  );
});
app.post("/students", (req, res) => {
  const {name, email, major} = req.body;
  db.query(
    "INSERT INTO students(name, email, major) VALUES(?, ?, ?)",
    [name, email, major],
    (err) => {
      if(err) throw err;
      res.json({message: "Student Added"});
    }
  );
});
app.put("/students/:id", (req, res) => {
  const {name, email, major} = req.body;
  db.query(
    "UPDATE students SET name=?, email=?, major=? WHERE id=?",
    [name, email, major, req.params.id],
    (err) => {
      if(err) throw err;
      res.json({message: "Student Updated"});
    }
  );
});
app.delete("/students/:id", (req, res) => {
  db.query(
    "DELETE FROM students WHERE id=?",
    [req.params.id],
    (err) => {
      if(err) throw err;
      res.json({message: "Student Deleted"});
    }
  );
});
app.listen(5000, () => {
  console.log("Server running on port 5000");
});

```

App.jsx:

```
import {useState,useEffect} from "react";
import axios from "axios";
function App(){
  const [students,setStudents]=useState([]);
  const [name,setName]=useState("");
  const [email,setEmail]=useState("");
  const [major,setMajor]=useState("");
  useEffect(()=>{
    loadStudents();
  },[]);
  const loadStudents=()=>{
    axios.get("http://localhost:5000/students")
    .then(res=>setStudents(res.data));
  }
  const addStudent=()=>{
    axios.post("http://localhost:5000/students",{
      name,email,major
    }).then(()=>{
      loadStudents();
    });
  }
  const deleteStudent=(id)=>{
    axios.delete(`http://localhost:5000/students/${id}`)
    .then(()=>{
      loadStudents();
    });
  }
  return(
    <div>
      <h1>Student Management System</h1>
      <input placeholder="Name" onChange={(e)=>setName(e.target.value)} />
      <input placeholder="Email" onChange={(e)=>setEmail(e.target.value)} />
      <input placeholder="Major" onChange={(e)=>setMajor(e.target.value)} />
      <button onClick={addStudent}> Add Student </button>
      <table border="1">
        <thead>
          <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Major</th>
            <th>Delete</th>
          </tr>
        </thead>
        <tbody>
          {
            students.map(student=>(
              <tr key={student.id}>
                <td>{student.id}</td> <td>{student.name}</td> <td>{student.email}</td>
                <td>{student.major}</td>
                <td>
                  <button onClick={()=>deleteStudent(student.id)}> Delete </button>
                </td>
              </tr>
            ))
          }
        </tbody>
      </table>
    </div>
  );
}
export default App;
```